

Semantic Binding Conversion Documentation

This directory documents the Structured-CSV-to-target-flat-file conversion program.

The semantic binding converter reads a Phase 1 UADC Structured CSV and a semantic binding CSV. It writes a Phase 2 target flat file such as ADS PSV or ISO 21378 ADC CSV. The binding table defines which target column receives which UADC **semantic_path** value, and which semantic class controls repeated target rows.

The ISO 21378:2019 ADC invoice binding currently covers Tables 38, 39, 53, and 54. The target TAX groups are expanded to four numbered column sets and BUSINESS SEGMENT is expanded to five numbered columns so that every emitted CSV header is unique. Each target row also records **mapping_status** and **mapping_note**. These columns distinguish direct mappings from semantic approximations, required transformations, and source data that is not available from an EN 16931 invoice.

These four views complete the ISO 21378 part of the planned Phase 2 PoC baseline. Completion means that target definitions, bindings, CSV generation, regression execution, and explicit gap classification are present. It does not assert full ISO 21378 data completeness from an EN 16931 source invoice.

For implementation-level details shared with the broader processing model, including internal **dict/list/dataclass** objects and function-level data flow, use **../README_SCRIPT_PROCESSING.md**.

Files

- **program_specification.md** - Defines converter inputs, outputs, semantic binding rows, repeated row scope, indexed repeated values, output naming, and non-goals.
- **user_guide.md** - Gives command examples for ADS PSV and CSV generation, directory input, and troubleshooting.
- **iso21378_adc_invoice_coverage.md** - Records ISO 21378 ADC invoice mapping coverage, data gaps, and the Phase 1 regression result.

Related Directories

- **../src/** - Converter implementation, especially **semantic_binding.py**.
- **../specs/bindings/semantic/** - Active semantic binding CSV files.
- **../specs/bindings/syntax/** - Syntax binding CSV files that define the Phase 1 Structured CSV model used as semantic source.
- **../out/phase1/** - Phase 1 Structured CSV inputs.
- **../out/phase2/** - Generated Phase 2 target outputs.

Phase 2 semantic binding starts from the common Structured CSV. It does not parse XML directly and does not depend on source XML XPath expressions.